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Business model innovation in SMEs: the role of boundaries in the digital era

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ABSTRACT

The digital era is radically changing our societies and how firms do business. Innovating the business model has become a fundamental capability to survive competition, particularly for small and medium-sized enterprises (SMEs). This study investigates the digital-era role of boundary management capabilities in the processes of Business Model Innovation (BMI) in SMEs. Structural Equation Modelling is adopted to analyse data from a survey of 250 Italian experts who possess direct research and consulting experience with SMEs. Our findings reveal that digitalisation and firms' boundaries affect BMI in SMEs. Moreover, our results demonstrate how boundary management, specifically its technological and relational aspects, directly impacts BMI and mediates the relationship between boundary size and BMI. The study also offers important theoretical and practical insights, calling on scholars and managers to give more attention to the boundary management of SMEs in order to support BMI.

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Digitalisation; SME; Business model; boundary capabilities

1. Introduction

Digital technologies are powerful forces pushing firms toward new business models (Cohen and Kietzmann 2014; Fjeldstad and Snow 2018), making the capability to innovate increasingly relevant (Chesbrough 2010; Foss and Saebi 2018; Zott and Amit 2007) and emphasising the shift from the boundaries of management to the management of boundaries (Caputo, Fiorentino, and Garzella 2018). Studies from economic, management and organisational behaviour perspectives have referred to the boundary concept to analyse resources, activities and processes that can be jointly controlled and influenced by many organisations (Yang, Lin, and Lin 2010). As such, boundaries are defined as transitional areas between the inside and the outside of an organisation, circumscribing resources and capabilities over which governance and control are extended (Fiorentino 2016). Boundaries should be viewed as a continuum; an area in which it is difficult to distinguish the firm from its external environment (Normann and Ramírez 1993). Therefore, it is increasingly necessary to consider the autonomy of the concept of boundaries as having intrinsic characteristics, such as the size of these boundaries, namely how much the continuum area is extended, the resources that are positioned in the boundaries, and their management (Caputo, Fiorentino, and Garzella 2018).

Existing research has predominantly focused on large firms, with limited studies exploring how inherent characteristics, an organizational-wide approach to collaborative innovation (Fjeldstad and Snow 2018) and unique contexts prevalent within SMEs can affect business model innovation (Anwar 2018; Child et al. 2017). Some authors highlight the fact that large firms are more active

than SMEs in leveraging external resources for innovation (Dooley, Kenny, and Cronin 2016). At the same time, digitalisation represent an opportunity for SMEs, where a less structured organisation can faster develop the capability to proactively lead Business Model Innovation (BMI) (Lindgren and Abdullah 2013). Digital technologies can increase levels of production efficiency to reduce production costs while optimising immobilised capital, by minimising inventories and streamlining information flows. Indeed, firms can limit risks by sharing processes and activities to reduce the risks of flow interruptions along the supply chain, minimising operational risks contingent on operating costs (Fiorentino 2016). The relevance of network models and of the development of a digital society exert pressures on the management of the business processes concerning firms boundaries (Caputo, Fiorentino, and Garzella 2018; Schotter et al. 2017).

Our research responds to the need of studying the BMI construct within the SMEs context, prevailing in many countries such as Italy, and aims to delineate a line of actions and solutions on how SMEs can develop innovative business models and foster boundaries capabilities management in the new digital landscape. Our aim is to analyse how SMEs should successfully implement more innovative and co-creational business models. Specifically, we formulate the following research question: how do firms boundaries affect business model innovation of SMEs in the digital era? Boundary management is a concept advanced in previous research (Caputo, Fiorentino, and Garzella 2018; Garzella 2000) and is considered a set of capabilities developed by managers and organisations in order to manage the web of resources and relationships in the boundary areas of a given firm. Boundary management capabilities are conceptualised into three categories: technological, relational and cultural capabilities (Caputo, Fiorentino, and Garzella 2018). We advance this theoretical model analysing the boundary management capabilities in contexts of rapid technological change, where the need for continuous BMI through cooperation, competition and a search for new markets is pivotal. Our model highlights how the size of boundaries and a firm's capabilities to manage them can be relevant to the elaboration and implementation of innovative business models in SMEs. We tested our hypotheses through a Structural Equation Model that analyses data collected by a survey submitted to professionals (e.g. CEOs, advisors and consultants) in Italian SMEs.

2. Theoretical background

Business model literature has successfully developed frameworks and practices to capture the complex interrelationship between the creation and appropriation of value and to understand the logic of an organisation for value creation (Massa, Tucci, and Afuah 2017; Zott, Amit, and Massa 2011). Stemming from prior accounts of business models, we use a comprehensive definition which sees business models as a modelling and representation tool. It allows businesses to mediate technological and other resources in several ways: controlling, communicating and innovating; classifying, disclosing, focusing managerial attention, helping idea exploration, and supporting and coordinating the knowledge flow (Zott, Amit, and Massa 2011). A business model can be made of an internal value chain: product and services, activities, resources; and of an external value chain: customers, partners, competitors. Business models help innovation since they turn market opportunities into profits, delivering the value of a service or product through commercialisation (Zott and Amit 2012). Companies commercialise new ideas and technologies through their business models; but while companies have extensive investments and processes for exploring new ideas and technologies, they often have little ability to innovate their business models (Chesbrough 2010; Spieth, Schneckenberg, and Matzler 2016).

In line with these assumptions, more companies now are turning toward BMI as an alternative or complement to product or process innovation and because it can translate to a sustainable advantage (Brehmer, Podoyntsyna, and Langerak 2018). The digital era and the Industry 4.0 paradigm, combining different technologies, open unforeseen possibilities and offer the potential not only to create radically new products and services and to share knowledge between different actors of the technology ecosystem (Lombardi 2019) but also to generate innovative business models

(Caputo, Marzi, and Pellegrini 2016; Spieth, Schneckenberg, and Matzler 2016). BMI refers to a new activity system of a firm (Foss and Saebi 2016) and innovative structures for value creation and value capture (Chesbrough 2010) involving a single firm and its alliance partners and customers (Bouncken and Fredrich 2016). Recent developments emphasise a need for a more dynamic perspective that addresses BMI itself so that any fundamental change in the relationship between the model elements can be understood as BMI (Foss and Saebi 2016).

Firms often experience conditions of resource scarcity, especially SMEs, and they need to focus on BMI as source of future value (Amit and Zott 2012). Competitors might find it more challenging to imitate or replicate an entire novel activity system than a single new product or process (Casadesus-Masanell and Zhu 2013). Managers must be aware of the possibility of competitors' efforts in this area because BMI can be a powerful competitive tool. Competitive pressures have pushed BMI much higher than expected on firms' priority lists. Even under conditions of resource scarcity, organisations do not need to renounce innovation as a way of enhancing their performance (Zott and Amit 2008). Instead, managers should consider the opportunity offered by BMI to complement innovation in processes and products. BMI can allow managers to resolve the apparent trade-off between innovation costs and benefits by addressing how they do business and by involving partners in new value-creating activity systems.

Prior studies have shown BMI as a core driver for a firm's survival and superior performance, especially in growing industries. However, the role of BMI has been discussed theoretically, while empirical studies still lack reference to SMEs (Child et al. 2017). Moreover, traditional definitions of business models considering internal or external value chains (Zott and Amit 2012) demonstrate that the literature focuses mostly on internal or external elements, giving less importance to the connecting elements: boundaries and how boundary management can be significant for the elaboration and implementation of a business model (Caputo, Fiorentino, and Garzella 2018).

Accordingly, we wish to shed light on the influence of digital technologies on firms' boundaries and, consequently, on BMI. Indeed, boundary management requires firms to develop the capability to manage resources that are neither internal nor external, with the need to balance new technological tools (e.g. cloud technologies, sensors, big data, 3D printing) and new inter-organisational relationships (e.g. cooperation, collaboration and communication between the firms' networks) together with the cultural evolution due to the integration of new ways of creating a sense of belonging as well as with new leadership styles (Caputo, Fiorentino, and Garzella 2018).

3. Hypotheses development

Digital technologies have radically changed the nature and structure of new products and services, shaped novel value creation and value appropriation pathways, enabled innovation collectives that involve dynamic sets of actors with diverse goals and capabilities and produced a new breed of innovation processes (Porter and Heppelmann 2015). Digitalisation creates (and consequently changes) market offerings, business processes and models that result from the use of new technology (Caputo, Fiorentino, and Garzella 2018; Caputo, Marzi, and Pellegrini 2016). Digital innovation management refers to the practices, processes, and principles that underlie the effective orchestration of digital innovation (Nambisan et al. 2017).

Digital technologies have changed the way of doing business, and the first reaction of scholars and operators has been that of linking new forms of strategic development to the concept of boundaries (Brouthers and Hennart 2007; Fiorentino 2016; Nambisan et al. 2017; Schotter et al. 2017). In fact, some authors argue that digital technologies dissolve firms' boundaries (Lindgren and Abdullah 2013) and shift agency of traditional entrepreneurship and innovation processes and outcomes (Nambisan et al. 2017). However, other scholars suggest that the new competitive context and digital convergence are able to develop intra and inter-organisational boundaries (Schotter et al. 2017; Yang, Lin, and Lin 2010). Phelps (2007) modelled the effects of the increasing utilisation of information technology on organisational boundaries resulting in a more heterogeneous network of

boundaries. Furthermore, Caputo, Fiorentino and Garzella (2018) theorised that the digital transformations of our time are creating new competitive contexts, and such contexts are having a major impact on firms and business processes, making the boundaries of the firms fluid, dynamically expanding and more blurred than in the past (Figure 1).

Thus, we expect that digital technologies impact firms' boundary sizes in a way that enlarges the dimensions and the relevance of the continuum area of the business' borders in which it is not easy to distinguish internal or external environments of firms. Accordingly, we formulate the following hypothesis:

Hypothesis 1: Digital technologies positively affect SMEs' boundary size.

Understanding new value creation and value appropriation pathways which enable innovation collectives to involve a dynamic set of actors with diverse goals and capabilities should force businesses to pay attention to their *boundary size*. In any case, this means shifting attention of firm leaders in the business process periphery. The management of boundaries is designed to create value by focusing on business processes and activities that occur at a firm's boundaries (Fiorentino 2016). In a context in which new communication tools and new ways of governing relationships are established, bringing boundaries to the centre of the strategy promotes the bearing or linking of strategies and can in turn promote creativity and innovation (Foss and Saebi 2018; Phelps 2007).

Managing boundaries involves a set of capabilities that guarantees balance and harmony between various elements and activities combined with internal and external forces (Parmigiani and Mitchell 2009). To manage effectively, it is necessary to invest in management personnel attitudes, a fact that highlights the importance of governing relational, organisational and technological factors (Caputo, Fiorentino, and Garzella 2018).

In line with the enlargement of boundary size, boundary management teams must express the increasingly urgent need to organise innovative relational systems (Conway 1997). According to Yan et al. (2019), top management behaviour has a significant impact on BMI. This organisational effort must extend beyond classical business boundaries and interpret new ways of managing processes that move resources from within to boundary areas (e.g. issues related to the development of telework or the 'internet of things') as well as processes that approach resources from the outside (e.g. the creation of inter-company networks or 'big data analytics'). The boundary manager emerges as a new figure in the business landscape, one who has to govern the flow of information and corporate communications both internally and externally. Moreover, he or she must commercially and

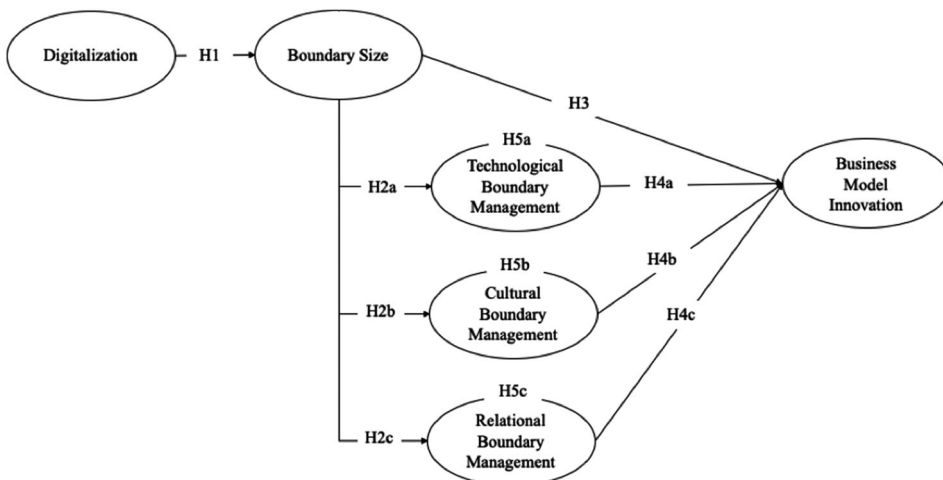


Figure 1. Theoretical model.

operationally support the company's mission and represent its values and culture while maintaining relationships with partners. To render the information system and the circulation of information effective, the boundary manager must organise the flow of information itself in a bidirectional manner to identify weak signals received from inside and outside (Phelps 2007). The need to deliver creativity and flexibility to the corporate system while maintaining stability implies the need to redefine interpersonal relationships, the role system, leadership style and control methods (Kickul and Gundry 2001).

Typical management and organisational issues relating to the enlargement of boundary size are all represented by the difficulty of controlling organisations and individuals gravitating to the boundary area over time and representing the strategically relevant resources (Berglund and Sandström 2013). Barner-Rasmussen et al. (2014) argue that while boundary enlargement involves linking external and internal groups, it is more profoundly about intervening in the process in such a way to mediate conflict, build competence and negotiate meaning across boundaries. By engaging in the process of boundary enlargement, selective managers produce new kinds of practices that serve to link individuals across boundaries (Birkinshaw, Ambos, and Bouquet 2017; Levina and Vaast 2017).

The change of firms' boundary size due to digitalisation allows the reduction of physical, temporal, relational and even cultural barriers (Garzella 2000). As conceptualised by Caputo and colleagues (2018), there are three relevant factors for the management of boundaries: technological, cultural and relational capabilities. Consequently, we formulate the following related hypothesis:

Hypothesis 2a: Boundary size positively affects SMEs technological boundary management capability.

Hypothesis 2b: Boundary size positively affects SMEs cultural boundary management capability.

Hypothesis 2c: Boundary size positively affects SMEs relational boundary management capability.

Considering the relations between boundaries and BMI, Bouncken and Fredrich (2016) analyse how firms' sizes, alliance experiences and the alliance durations influence value capture from BMI. What they demonstrate highlights the role of openness and the role of users in influencing the development of technologies and business model choice (Baden-fuller and Haefliger 2013; Foss and Saebi 2018).

Openness refers to the permeability of a company's boundaries. This drives firms towards the use of new 'sharing economy' business models (Cohen and Kietzmann 2014). In a global environment where the digital revolution has substantially enhanced the rate of competition, cooperation strategies applied within company departments and between companies can often have a number of benefits, including improved levels of communication, stronger relationships between suppliers and customers and process sharing (e.g. co-design and co-production) enlarging firms' boundaries size. Moreover, other authors (Dooley, Kenny, and Cronin 2016) highlight that large-scale firms are more active than SMEs in leveraging external resources for innovation and that small-sized firms are least active in this practice.

This scenario is significant when, due to a lack of knowhow or funds, businesses can hardly self-generate the innovation needed to effectively respond to market needs. SMEs face many more fundamental and basic challenges on their digital transformation path due to a scarcity of funds. Therefore, only through expanding the boundary size and collaborating with suppliers and customers can they achieve BMI. Our third hypothesis is shaped by the above considerations:

Hypothesis 3: Boundary size positively affects SMEs business model innovation.

Teece (2018) argues that business models are more context-dependent than technology, depending on resources and capabilities that are available within the respective company. Therefore, the role of management and managers emerges as a relevant factor in BMI and boundary management. Boundary managers should develop a great capability to manage boundary resources and the relationship between the firm and the environment, using all relational, communicational,

technological and technical attitudes (Kickul and Gundry 2001; Takeishi 2001). Accordingly, we formulate the following hypothesis:

Hypothesis 4a: Technological boundary management capability positively affects SMEs business model innovation.

Hypothesis 4b: Cultural boundary management capability positively affects SMEs business model innovation.

Hypothesis 4c: Relational boundary management capability positively affects SMEs business model innovation.

Having established that boundary size is connected to boundary management capability and, in view of the positive influence of the boundary management capability on BMI, we also hypothesise that:

Hypothesis 5a. Technological boundary management capability mediates the positive relationship between boundary size and business model innovation.

Hypothesis 5b. Cultural boundary management capability mediates the positive relationship between boundary size and business model innovation.

Hypothesis 5c. Relational boundary management capability mediates the positive relationship between boundary size and business model innovation.

4. Methods

Procedure and sample

We administered an online questionnaire to a sample set of Italian professionals who possess direct expertise in managing and/or consulting with SMEs about strategic decisions. The questionnaire was back-translated, tested by a panel of experts and piloted with 20 other experts to ensure appropriateness within the Italian context. Survey administration took place in late 2018 through the members list of the Italian Association of Business Administration and Management (*AIDEA – Accademia Italiana di Economia Aziendale*), which comprises 2,287 academics in business administration, accounting, management, marketing and organisation in Italy. Most of the AIDEA members are administrators, CEOs, advisors or consultants in both Italian and foreign companies. Only those with an active managerial or consulting role in one or more SME were included in the sample. Thus, our survey targeted experts with both business and academic experience who have a holistic view of the changes and challenges facing SMEs.

A total of 439 questionnaires were filled, however, 189 of these were incomplete. Therefore, we used 250 completed questionnaires in our data analysis (this represents an 11% response rate). To limit the possible bias in self-administered surveys due to directional responses, we performed a series of robustness checks to assure that bias did not jeopardise the validity of our data (Podsakoff et al. 2003). In particular, we conducted a *t*-test comparing different categories of respondents, and we checked for common method variance using the Harman one-factor method (Caputo, Ayoko, and Amoo 2018; Podsakoff et al. 2003). All tests delivered satisfactory results.

Measures

All scales used in the questionnaire were derived from existing literature, adopting 5-point Likert scales ranging from 1 ('strongly disagree') to 5 ('strongly agree'). We measured *digitalisation*, i.e. the relevance of digital technologies for SMEs, through a 4-item scale adapted from previous studies (Caputo, Fiorentino, and Garzella 2018; Kuester, Konya-Baumbach, and Schuhmacher 2018; Tarutè and Gatautis 2014). Respondents were asked to indicate the extent to which they agreed with different statements concerning the impact of digital technologies (e.g. IoT, AI, Big Data, Social Media). *Boundary size* was measured with a scale adapted from Garzella (Garzella 2000) and Galeotti and Garzella (2013). Respondents were asked to indicate the extent to

which the boundary area of SMEs had increased in size. The construct of *boundary management capability* was operationalised through three first-order constructs. To measure *relational boundary management* (RBM), *cultural boundary management* (CBM) and *technological boundary management* (TBM), we used a scale adapted from Caputo, Fiorentino, and Garzella (2018). To measure *BMI*, we used a scale adapted from previous studies (Snihur and Wiklund 2018; Zott and Amit 2007).

Reliability and validity

To determine the reliability (i.e. internal consistency) and validity of measurements, we adopted the AMOS v25 package in SPSS. Following Hair et al. (2016), we assessed internal consistency using Exploratory Factor Analysis, Cronbach's Alpha, Average Variance Explained (AVE) and Composite Reliability (CR). Generally, factor loadings above 0.50, Cronbach's alphas exceeding 0.70, AVEs above 0.50 and CR values above 0.70 are considered satisfactory (Fornell and Larcker 1981; Nunnally and Bernstein 1994). All our constructs were first-order, and as shown in Table 1, all returned satisfactory measures. As evidence of convergent validity, we have all loadings above 0.50. As evidence of discriminant validity, we found no strong cross loadings above 0.30 and we report no correlations between factors above 0.60.

To further assure the validity of our multi-dimensional constructs, we also conducted Confirmatory Factor Analyses (CFA) using maximum likelihood estimation. Following the most recent indications in the literature (Caputo et al. 2019), we ran a number of tests for model fit that show that our measurement model meets the acceptable threshold for all tests. In particular, our SRMR is 0.0533, our RMSEA is 0.055, our PCLOSE is 0.139, our CIF is 0.997 and our CMIN/DF is 1.766.

In order to check for multicollinearity issues among our variables, we followed Hair et al. (2016) and examined whether there are bivariate correlations above 0.90 or several bivariate correlations

Table 1. Evaluation of construct.

Latent Variable	Item	Factor						Cronbach's Alpha	C.R.	AVE
		1	2	3	4	5	6			
Digitalization (DIGI)	DIGI01				0.82			0.8	0.8	0.51
	DIGI02				0.83					
	DIGI03				0.61					
	DIGI04				0.57					
Boundary Size (BS)	BS01					0.8		0.82	0.8	0.59
	BS02					0.79				
	BS03					0.72				
Relational Boundary Management (RBM)	RBM01						0.59	0.75	0.7	0.59
	RBM02						0.91			
Cultural Boundary Management (CBM)	CBM01	0.71						0.89	0.9	0.53
	CBM02	0.68								
	CBM03	0.71								
	CBM04	0.78								
	CBM05	0.74								
	CBM06	0.79								
	CBM07	0.68								
Technological Boundary Management (TBM)	TBM01			0.62				0.88	0.9	0.64
	TBM02			0.57						
	TBM03			1.01						
	TBM04			0.92						
Business Model Innovation (BMI)	BMI01	0.6						0.88	0.9	0.57
	BMI02	0.65								
	BMI03	0.9								
	BMI04	0.81								
	BMI05	0.77								

Notes: Extraction method for the factor loadings used was Maximum Likelihood, Promax rotation with Kaiser normalization. KMO measure of sampling adequacy 0.882. Bartlett's test of sphericity Chi-square 3444.251, df 300, $p < 0.000$. Goodness-of-fit Test Chi-Square 317.844, df 165, $p < 0.000$. All communalities values were below 0.350. Total variance explained 60.10%.

above 0.70. The highest correlation among our variables is 0.616, as seen in the correlation matrix in Table 2. Thus, we concluded that there are no multicollinearity problems.

5. Results

We tested our hypotheses through a Structural Equation Model based on our theoretical model with SPSS AMOS v25. Prior to this test, we tested for outliers through a Cook's distance analysis, and we did not find influential outliers in our set of predictor variables (all below 0.05). Two popular methods used to detect mediation are the Sobel test (Sobel 1982) and the causal approach utilised by Baron and Kenny (1986). However, the Sobel test has often been criticised because it requires that the sampling distribution of the indirect effect should be normal, while the causal approach has been criticised because of low statistical power (Fritz and MacKinnon 2007; Hayes 2009). We followed the indication to use bootstrapping (Hayes 2009), especially when the sample is large enough ($n=250$), reducing the risk of committing a type 1 error, which is a criticism of this technique (Koopman et al. 2015).

We undertook our mediation test using the AMOS SPSS v25, using 2,000 bootstrap samples, maximum likelihood estimators, a bootfactor of 1 and 95% bias-corrected confidence intervals. Overall, this structural mediation model explains the data well ($X^2 = 10.274$, $X^2/df = 3.425$, $p = 0.016$, CFI = 0.987, RMSEA = 0.099, SRMR = 0.048). Digitalisation explains the medium level of variance in boundary size ($R^2 = 0.125$), which in turn explains medium levels of variance in boundary management variables (TBM, $R^2 = 0.245$; CBM, $R^2 = 0.166$; RBM, $R^2 = 0.113$). Altogether, they explain the relatively high variance in BMI ($R^2 = 0.566$) (Table 3).

Our results indicate significant direct effects of digitalisation on boundary size (H1 – 0.354***), of boundary size on technological (H2a – 0.430***), cultural (H2b – 0.407***), relational boundary management (H2c – 0.335***), and BMI (H3 – 0.414***). Therefore, our data supports hypotheses 1, 2a, 2b, 2c and 3. Our analysis also shows that technological (H4a – 0.275***) and relational boundary management (H4c = 0.223***) have significant direct effects on BMI, while cultural boundary management (H4b) does not demonstrate a significant effect. Therefore, we reject hypothesis 4b while supporting hypotheses 4a and 4c.

Regarding indirect effects, we tested mediation of TBM, CBM and RBM in the relationship between boundary size and BMI. Our results show that TBM and RBM mediate this relationship, with an effect of 0.114 ($p = 0.000$) and 0.072 ($p = 0.000$) respectively, while the indirect effect of CBM was not significant. We also did a post-hoc power analysis which showed that our analysis has very good statistical power (Observed statistical power = 1.0, $p = 0.05$) to detect existing significant effects. Hence, we are confident that the insignificant effect of CBM on BMI is truly insignificant. Our results for the indirect effects support hypotheses 5a and 5c and reject hypothesis 5b.

6. Discussion

The results of this study confirm the proposed theoretical model which links competitive dynamics and trends deriving from digitalisation with the changes of SMEs boundaries' size and management

Table 2. Descriptive statistics.

	Mean	S.D.	CR	AVE	MSV	MaxR(H)	BS	CBM	BMI	TBM	Digi	RBM
BS	2.66	0.67	0.82	0.61	0.33	0.824	0.78					
CBM	2.35	0.49	0.89	0.53	0.32	0.901	0.36	0.73				
BMI	2.64	0.65	0.88	0.6	0.33	0.893	0.57	0.46	0.78			
TBM	2.48	0.7	0.88	0.66	0.31	0.934	0.44	0.49	0.56	0.81		
Digi	3.37	0.6	0.81	0.52	0.12	0.831	0.3	0.22	0.3	0.35	0.72	
RBM	3.26	0.72	0.75	0.6	0.32	0.752	0.27	0.57	0.44	0.39	0.24	0.77

Notes: Square-root of AVEs are in the diagonals.

Table 3. Structural model assessment.

Constructs	R^2		
Boundary Size (BS)	0.13		
Technological Boundary Management (TBM)	0.25		
Cultural Boundary Management (CBM)	0.17		
Relational Boundary Management (RBM)	0.11		
Business Model Innovation (BMI)	0.57		
Relation	Beta	<i>t</i> values	<i>p</i> values
Digi – BS	0.35	5.966	0.001
BS – TBM	0.43	7.581	0.001
BS – CBM	0.41	7.039	0.001
BS – RBM	0.34	5.618	0.001
BS – BMI	0.41	8.402	0.001
TBM – BMI	0.28	5.227	0.001
CBM – BMI	0.04	0.735	0.462
RBM – BMI	0.22	3.99	0.001

(Berglund and Sandström 2013; Cohen and Kietzmann 2014; Levina and Vaast 2017). Indeed, the findings demonstrate how interaction between the analysed variables affects BMI, and thus contribute to a better understanding of the main role of boundary management capabilities for BMI in SMEs.

The defined theoretical model tested confirms most of the research hypotheses. The results supporting H1, indicating that digitalisation affects the companies' boundary size, allow us to assert that also for SMEs the digital technologies shift attention to new forms of strategic development in which boundaries are the site of new value creation and value appropriation pathways. In particular, as the literature suggests, new competitive contexts and digital convergence are able to develop intra and inter-organisational boundaries that shift the attention of firms to the business processes periphery (Porter and Heppelmann 2015; Schotter et al. 2017; van Gils and Rutjes 2017). This result supports the concept that digital innovations increase firms' boundary size; therefore, our findings suggest that digital technology developments do not make boundaries disappear, rather they change and increase boundaries' relevance (Caputo, Fiorentino, and Garzella 2018; Garzella 2000).

The results supporting H2 demonstrate that the boundary size of SMEs affects technological, cultural and relational management capabilities. In this perspective, we confirm the argument of previous studies which claims that managing boundaries requires a set of capabilities that guarantees balance and harmony between various internal and external forces and resources (Fiorentino 2016; Schotter et al. 2017). To manage effectively, it is necessary to invest in personnel management attitudes, thus highlighting the importance of governing relational, cultural and technological factors.

The results supporting H3 validate that, in the context of SMEs, boundary size positively affects BMI. Some authors have already stated that a firm's size is a variable influencing value capture from business models (Bouncken and Friedrich 2016; Dooley, Kenny, and Cronin 2016). We add that boundary size enlargement is an element enhancing the permeability of firms' boundaries and pushing SMEs towards the implementation of innovative business models (Anwar 2018; Dilger 2016).

Moreover, results related to H4 pertain to the role of management and boundary managers in developing BMI. According to the literature, what emerges is the relevance of using all relational, communicational, technological and technical capabilities that we can consider as dynamic capabilities at the boundary level (Teece, 2018). This scenario is significant when SMEs cannot self-generate the innovation needed to effectively respond to market needs, often due to a lack of funds. Additionally, through the results of H5, we assert that technological and relational management capabilities have a mediating role in affecting the correlation between SMEs' boundary size and BMI.

On the contrary, cultural boundary management (CBM) capability is not confirmed as positively affecting BMI. The results of our empirical analysis reject H4b and H5b, which consider the relationships between CBM capability and SMEs' BMI. These results regarding the mediating role between

boundary size and BMI deserve an in-depth discussion as they are quite counterintuitive. Digitalisation forces are found to be positively related to CBM capability; hence, such pressures help CBM development. However, the cultural aspect of boundary management does not play any role in supporting the innovation of the business model. Considering the determinants of BMI, the changes in boundary size and the expansion of the boundary area have been confirmed as relevant, together with the push of digitalisation. In this context, only relational and technological capabilities prove relevant, while cultural variables – such as entrepreneurship, creativity, leadership – are not significantly correlated to boundary size and BMI.

We note how the specific focus of the research relates to the understanding of why the hypotheses related to cultural capability are rejected. This could be explained by the specific context of SMEs. Digitalisation has led to an expansion and a review of firms' boundaries such as to impose a rethinking of the business model. Technological and relation boundary management capabilities are relevant, while factors such as entrepreneurship, creativity, and leadership skills (CBM capabilities) do not seem to impact BMI. Nevertheless, it could be possible to argue that, unlike prior studies, SMEs face BMI issues by taking the 'starting' input from elsewhere. For this reason, SMEs should attend to relational and technological boundary capabilities, in order to be linked to the 'real' business model innovators more than to entrepreneurship, creativity and innovation. Therefore, most SMEs might transpose, based on relational boundaries and technological skills, the changes in the business models that larger firms have introduced. An alternative interpretation of the findings showing a lack of correlation between CBM and BMI could argue that management of CBM capabilities is less relevant for BMI due to a lack of internal relational and technological capabilities in SMEs on the one hand, and due to internal cultural capabilities are already well grounded in SMEs on the other.

7. Conclusions

This paper enriches the literature on BMI, innovation in SMEs and boundary management by confirming the relevance of boundary size and capabilities to the elaboration and implementation of innovative BMs in the SMEs context. Firstly, our study contributes to the growing literature on boundary management within the field of business models by presenting one of the first empirical research projects on this topic. Furthermore, this study contributes to the body of knowledge on innovation in SMEs by providing new insights on the differences between SMEs and large organisations. Finally, our project opens the door for future studies to deepen the investigation into the role of CBM capabilities. In particular, future scholars could test arguments that claim CBM capability is not relevant for BMI in SMEs.

This study also offers important practical implications. Our findings should push SMEs to give even more attention to how boundaries are impacting their business models currently and in the future. SMEs can make central in their strategy the management of the boundaries and the development of boundary management capabilities, also by exploiting the opportunities offered by digital technologies. Therefore, enlargement of boundary size affects the organisation of the supply chain and results in new ways of creation and appropriation of value. Moreover, as outputs, SMEs obtain also the reduction of risks and funding issues because the enhancement of boundary management capabilities amplifies the creative process and leads businesses to overcome fundamental challenges on their digital transformation implementing innovative business models.

One limitation of our study is that it is a single country study of Italy. Nevertheless, our sample size is robust, the response rate is 11%, and the respondents are professionals also engaged in international debate. Another limitation of our research is that our study is cross-sectional; future studies should examine a longitudinal effect of boundary management capabilities on BMI.

In sum, our study demonstrates a close correlation in changes and the expansion of boundary size with digitalisation and the implementation of innovative business models. Boundary management stands out as a relevant issue in current competitive contexts, highlighting the need for a paradigm

shift in regards to the management of resources which neither internal nor external and can be physiologically placed in the boundary area.

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